

Self-calibration [HITACHI]

Theoretical background

In a digital measurement system that captures signals from a physical sensor using a 16-bit ADC, due to environmental effects, the measured signal does not directly represent the true physical signal. To achieve higher quality, signal needs to be filtered before being passed to the computing system.

Goal

The measured signal `i_data[0]` is affected by a **systematic, time-varying distortion** that depends on the auxiliary signal `i_data[1]`.

This distortion:

- Is deterministic (repeatable under the same conditions)
- Changes as the auxiliary signal changes
- Alters the observed signal in a consistent but unknown manner

As a result, the measured signal does not directly reflect the true physical signal. Your task is to design a system that reconstructs an estimate of the original signal $M[n]$ using only the available inputs. The goal is to recover a signal that matches the true underlying signal.

Provided module

You can edit only the `task_2.vhd` file. If you wish, you may add other files for supportive submodules.

The module has the following inputs and outputs:

Table 1 List of inputs and outputs of the `task_2` module.

Signal name	Signal type	Bit length
<code>i_clk</code>	Input	1 bit
<code>i_valid</code>	Input	1 bit
<code>i_rst</code>	Input	1 bit
<code>i_data[0]</code>	Input	16 bits
<code>i_data[1]</code>	Input	16 bits
<code>i_last</code>	Input	1 bit
<code>i_first</code>	Input	1 bit
<code>o_data</code>	Output	32 bits
<code>o_valid</code>	Output	1 bit
<code>o_last</code>	Output	1 bit

The reset is synchronous and active high.

Input and output interfaces

Two synchronous signals are available:

- Measured signal $i_data[0] \in [-32768, 32767]$ (signed 16-bit, 2's complement)
- Temperature value $i_data[1] \in [-32768, 32767]$ (signed 16-bit, 2's complement)

Both signals are sampled at the same rate and time.

In one packet task receives 2048 samples of input data, 16 bits each. The last sample in a packet is indicated by signal i_last . The task should send 1x 32 bit header + 2048x output samples, 32 bits each, and should indicate the last sample in the output packet by providing signal o_last .

The header shall indicate the arithmetic format of the output data:

- 0x01006666 Floating point
- 0x01007171 Fixed point (Q1.15)

Task receives data in the way shown in the figure below:

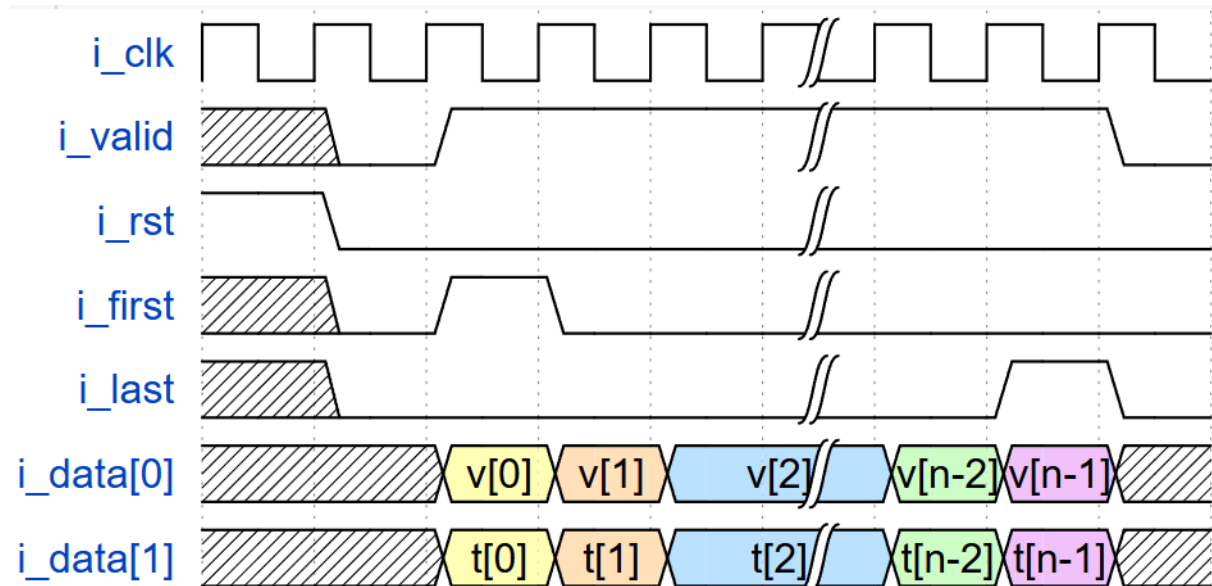


Figure 2 Input waveforms.

To send data from the task in a proper way one needs to provide the *o_valid* signal that is synchronized with data on the *o_data* bus and *o_last* signal that is synchronized with the last sample in output packet.

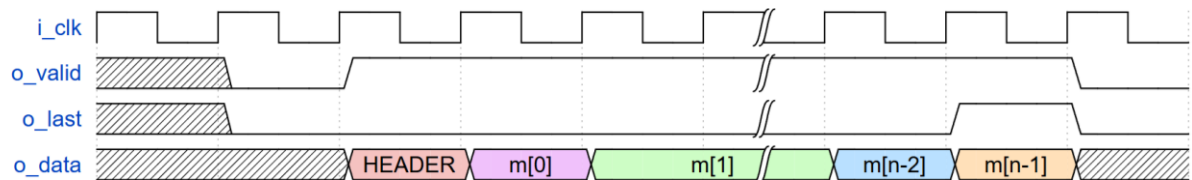


Figure 3 Output.

Output data can be sent as a continuous stream, or it can have regular or irregular gaps. The only requirement is providing *o_valid* and *o_last* signals.

Evaluating the task

Your solution will be evaluated by comparing your reconstructed signal with a reference signal.

The primary metric is Mean Squared Error (MSE): $\text{MSE} = (1/N) * \sum (M[n] - X[n])^2$

Lower MSE indicates better reconstruction quality.

For each of 2 golden vectors up to 15 points can be assigned:

- 15 points if $MSE < 3 * 10^{-11}$
- 10 points if $MSE < 3 * 10^{-7}$
- 5 points if $MSE < 3 * 10^{-3}$

Additional points up to 10 may be awarded for resource utilization if latency is smaller than 55ms. For more information, check chapter **2.3.2 Bonus Points in the FPGA_Hackathon_2026_Project_Guide** document).